6-4 Digital Representation of Characters

Point!

Digital Representation of Characters

(1) ([1]**Character code**) : A unique numerical value assigned to each character, symbol, etc.

(2) ([2]**Character code system**) : Summarizes the correspondence between characters and their respective character codes.

[1] ([3]**ASCII code**) : Only alphanumeric characters, symbols, and control characters (symbols used to control the computer). No Japanese characters (*kanji*, *hiragana*, or *katakana*).

		Most Significant 4 Bits								
		Binary	0000	0001	0010	0011	0100	0101	0110	0111
Least Significant 4 Bits	Binary	Hexadecimal	0	1	2	3	4	5	6	7
	0000	0	Symbols for Controlling Computers (Omission)		SP	0	@	P	`	p
	0001	1			!	1	A	Q	a	q
	0010	2			”	2	B	R	b	r
	0011	3			#	3	C	S	c	s
	0100	4			$	4	D	T	d	t
	0101	5			%	5	E	U	e	u
	0110	6			&	6	F	V	f	v
	0111	7			'	7	G	W	g	w
	1000	8			(	8	H	X	h	x
	1001	9			)	9	I	Y	i	y
	1010	A			*	:	J	Z	j	z
	1011	B			+	;	K	[	k	{
	1100	C			,	<	L	¥	l	|
	1101	D			-	=	M	]	m	}
	1110	E			.	>	N	^	n	~
	1111	F			/	?	O	_	o	DEL

<Example> Convert the string “Hello” into a binary character code and a hexadecimal character code. Combine the most significant 4 bits and the least significant 4 bits.

String	H	e	l	l	o
Character code (binary)	01001000	01100101	01101100	01101100	01101111
Character Code (hexadecimal)	48	65	6C	6C	6F

[2] ([7]**Unicode**) : A character code standard that consolidates characters from around the world into a single character code. Due to differences in code assignments, there are variations such as UTF-8 and UTF-16.

(3) ([8]**Encoding**) : Representing a string with character codes. The opposite is known as ([9]**decoding**).

(4) ([10]**Character corruption**) : A phenomenon that occurs due to mismatched encoding and decoding methods.

(5) ([11]**Font**) : The shape data of characters corresponding to character codes.
<Examples> Sans-serif, Serif, Semi-cursive, etc.

- To display characters on a computer screen or printer output, two elements are required: ([12]**character code**) and ([13]**font**).

Warm Up

Answer the following questions.

(1) Choose the numbers or terms that best fit into the blanks [1] to [5] from the options **A** to **H** below.

> Characters and symbols can also be represented by combinations of 0s and 1s. A system that maps characters and symbols to binary or hexadecimal is called a ([1]). There are several types within the ([1]), including ([2]), which accommodates the writing systems of countries around the world.
>
> Alphanumeric characters and symbols are represented by ([3]) byte, 1 byte can represent ([4]) types of characters.

A 1 **B** 2 **C** 128 **D** 256

E Character code **F** Character corruption **G** Unicode **H** ASCII code

(2) Use the character code table on the right to answer the following questions.

[1] Use binary to represent the character code for "M".

[2] Use hexadecimal to represent the character code for "MILK".

[3] What is the character string corresponding to the character code "$4C6F7665_{(16)}$"?

		Most Significant 4 Bits								
		Binary	0000	0001	0010	0011	0100	0101	0110	0111
Least Significant 4 Bits	Binary	Hexadecimal	0	1	2	3	4	5	6	7
	0000	0	Symbols for Controlling Computers (Omission)		SP	0	@	P	`	p
	0001	1			!	1	A	Q	a	q
	0010	2			"	2	B	R	b	r
	0011	3			#	3	C	S	c	s
	0100	4			$	4	D	T	d	t
	0101	5			%	5	E	U	e	u
	0110	6			&	6	F	V	f	v
	0111	7			'	7	G	W	g	w
	1000	8			(	8	H	X	h	x
	1001	9			)	9	I	Y	i	y
	1010	A			*	:	J	Z	j	z
	1011	B			+	;	K	[	k	{
	1100	C			,	<	L	¥	l	|
	1101	D			-	=	M	]	m	}
	1110	E			.	>	N	^	n	~
	1111	F			/	?	O	_	o	DEL

Explanation

(1) [1] **E** [2] **G** [3] **A** [4] **B** [5] **D**

(2) [1] From the character code table, the most significant 4 bits of "M" are 0100 and the least significant 4 bits are 1101; therefore, $01001101_{(2)}$

[2] According to the character code table, the character codes corresponding to "M, I, L, K" are 4D, 49, 4C, and 4B, respectively, in the order of most significant to least significant. Connecting these character codes results in $4D494C4B_{(16)}$

[3] Consider the character code as separated into "4C/6F/76/65".
"4C" corresponds to the character "L" because it is located at the intersection of the most significant 4 and the least significant C.
Similarly, "6F" corresponds to "o", "76" to "v", and "65" is "e", so the resulting character string is Love

Try

1 Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] to [4] with the appropriate terms and numbers.

> The rule for how characters and symbols are represented in binary or hexadecimal is called a (　[1]　). Alphanumeric characters and symbols are represented by (　[2]　) byte. In principle, 1 byte can represent (　[4]　) types of characters, and 2 bytes can represent 65,536 types of characters.

(2) Briefly explain the characteristics of the character code system Unicode.

(3) Use the character code table below to answer the following questions.

		Most Significant 4 Bits								
		Binary	0000	0001	0010	0011	0100	0101	0110	0111
Least Significant 4 Bits	Binary	Hexadecimal	0	1	2	3	4	5	6	7
	0000	0	Symbols for Controlling Computers (Omission)		SP	0	@	P	`	p
	0001	1			!	1	A	Q	a	q
	0010	2			”	2	B	R	b	r
	0011	3			#	3	C	S	c	s
	0100	4			$	4	D	T	d	t
	0101	5			%	5	E	U	e	u
	0110	6			&	6	F	V	f	v
	0111	7			'	7	G	W	g	w
	1000	8			(	8	H	X	h	x
	1001	9			)	9	I	Y	i	y
	1010	A			*	:	J	Z	j	z
	1011	B			+	;	K	[	k	{
	1100	C			,	<	L	¥	l	|
	1101	D			-	=	M	]	m	}
	1110	E			.	>	N	^	n	~
	1111	F			/	?	O	_	o	DEL

[1] What is the character code for “a” in binary?

[2] What is the symbol corresponding to the character code “$00111110_{(2)}$”?

[3] What is the symbol corresponding to the character code “$5C_{(16)}$”?

[4] What is the character string corresponding to the character code “$486172696E657A756D69_{(16)}$”?

(4) Calculate the following data size.

[1] The data size when “Welcome” is entered.

(5) Choose the shape data of characters corresponding to character codes from the options **A** to **C** below.

A Font　　**B** Environment-dependent characters　　**C** *Emoji*

Exercise

1 Answer the following questions.

(1) Answer the following questions.

[1] What is the term for a unique numerical value assigned to each character, symbol, etc.?

[2] How many types of characters can be represented with 1 byte?

[3] What is the term for representing a string with character codes?

[4] What is the term for a design that corresponds to a character code and shares the same characteristics?

(2) Briefly explain the characteristics of the character code system ASCII.

(3) Use the character code table below to answer the following questions.

		Most Significant 4 Bits								
		Binary	0000	0001	0010	0011	0100	0101	0110	0111
Least Significant 4 Bits	Binary	Hexadecimal	0	1	2	3	4	5	6	7
	0000	0	Symbols for Controlling Computers (Omission)		SP	0	@	P	`	p
	0001	1			!	1	A	Q	a	q
	0010	2			”	2	B	R	b	r
	0011	3			#	3	C	S	c	s
	0100	4			$	4	D	T	d	t
	0101	5			%	5	E	U	e	u
	0110	6			&	6	F	V	f	v
	0111	7			'	7	G	W	g	w
	1000	8			(	8	H	X	h	x
	1001	9			)	9	I	Y	i	y
	1010	A			*	:	J	Z	j	z
	1011	B			+	;	K	[	k	{
	1100	C			,	<	L	¥	l	|
	1101	D			-	=	M	]	m	}
	1110	E			.	>	N	^	n	~
	1111	F			/	?	O	_	o	DEL

[1] What is the character code for “Y” in binary?

[2] What is the character corresponding to the character code “$01101111_{(2)}$”?

[3] What is the character corresponding to the character code “$72_{(16)}$”?

[4] What is the character string corresponding to the character code “$5468616E6B596F7521_{(16)}$”?

(4) Calculate the following data size.

[1] The data size when “HappyBirthday” is entered